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ORIGINAL ARTICLE

The lobbying disclosure act at 25: Challenges and opportunities for analysis

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Abstract

The Lobbying Disclosure Act (LDA) will turn 25 years old on December 19, 2020. Between 1998 and 2018, organizations and lobbying firms filed more than one million reports. These reports provide information on the activity of 47,555 organizations, 6834 lobbying firms, and 47,479 individual lobbyists. The interest group scholarly community has put the LDA data to good use over the course of a quarter century of observation. Yet, lobbying disclosures have not been fully leveraged to generate new discoveries about the politics of interest groups, advocacy, legislative studies, and public policy. In this article, we provide an overview of the lobbying disclosure process by describing the idiosyncrasies of the data's nested and complex structure and the limitations of what information is actually reported. We then assess recent research using LDA data, provide advice on completing research at different levels of analysis, document important changes in data reliability and structure over time, and introduce replication tools to support further research. As the time horizon of lobbying disclosure panel data continues to grow, the LDA—especially when combined with other sources—is an increasingly fruitful way to observe money in politics, legislative behavior, corporate political activity, political consulting, and individual lobbyists engaged in policy advocacy.

Keywords Lobbying · Interest groups · Lobbying Disclosure Act · Honest Leadership and Open Government Act · Policy Agendas Project · Compustat

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The Lobbying Disclosure Act (LDA) of 1995[1] represents a watershed moment for the study of interest group politics in the United States. For the first time, the law effectively compelled organized interests to disclose how much they spend on lobbying activities, who their lobbyists are, what government institutions they contact, what general policy areas they advocate, and what specific issues on which they lobby government at the US national level. Additionally, the law requires lobbyists who previously worked in government to publicly disclose their previous federal employment, thus codifying transparency about the revolving door between the government and the private sector into law.

Prior to the LDA, the system was a shambles. For all practical purposes, the disclosure regime under the Regulation of Lobbying Act (RLA) of 1946 was merely a voluntary system that only a select few major lobbying firms used, most likely as a way to promote their business. The RLA was useless for interest group scholars, who instead tended to use commercial directories such as Congressional Quarterly's *Washington Information Directory* (e.g., Walker 1983) or *Washington Representatives* (e.g., Schlozman 1984; Schlozman and Tierney 1983, 1986; Heinz et al. 1993). These directories tended to be limited to information identifying an organized interest, selected lobbyists, and public affairs employees. They do not document any actual lobbying behavior or advocacy activities. Simply, how much the LDA improved the public's knowledge of what interests were seeking to influence public policy, what agents were hired to influence government, what issues they focused on, and to whom they were directing their advocacy cannot be overstated.

Incidentally, this new data treasure trove became available just as leading interest group scholars were calling on the field to produce more generalizable, empirically driven research that accounted for the role of political institutions and economic contexts across various jurisdictions and policy domains (Gray and Lowery 1996; Baumgartner and Leech 1998). Baumgartner and Leech (2001) report on the first-ever dataset constructed from the LDA (see also Baumgartner and Leech 1999). Since the initial reports in 1996 were filed on paper, this project consisted of painstakingly hand-coding information by a team of supervised undergraduate students from 19,000 original files stored on microfiche. The article, which has been cited more than 500 times at the time of writing, describes the now familiar extreme value distribution of many balkanized issue niches versus a select handful of issues that attract the lion's share of attention from organized interests. These findings were only possible because the LDA permitted the authors to account for the entire population of organized interests at the federal level, not merely those consulting firms that voluntarily disclosed or associations that advertised in the lobbyist phonebook.

Much has changed since the publication of "Interest Niches and Policy Bandwagons." In 1998, the Secretary of the Senate began allowing filers to submit forms electronically, and in 2004, it began requiring electronic submission. Soon after in

[1] The original LDA was amended by the 2007 Honest Leadership and Open Government Act (HLOGA), Unless otherwise noted, we refer to the LDA as amended (2 USC. Section 1601 et. seq.) in general. All technical definitions may be found in "Lobbying Disclosure Act Guidance," published by the Clerk of the House at https://lobbyingdisclosure.house.gov/ldaguidance.pdf.

2005, the nonprofit Center for Responsive Politics (CRP) began publishing lobbying disclosure data on *OpenSecrets.org*, a website long dedicated to publishing Federal Election Commission campaign finance information. The *OpenSecrets.org* Lobbying Database significantly improves the raw government data by cleaning and reorganizing it, resolving organization and lobbyist name ambiguities, and adding information about organization's primary economic activity. Academic research across disciplines that use lobbying disclosure reports skyrocketed once *OpenSecrets.org* data became available. We identify at least 85 peer-reviewed articles, chapters, and books that use LDA data as a primary source of empirical observation since the publication of Baumgartner and Leech (2001).

Though the LDA's role in fostering large-scale empirical research on interest groups and policy advocacy is remarkable, the lobbying disclosure regime and *OpenSecrets.org* data are still limited in many fundamental ways. There are a number of idiosyncrasies of the lobbying disclosure database that are not apparent from publicly available documentation from SOPR or *OpenSecrets.org*. Here, we describe the non-intuitive nested, complex structure of the lobbying database, report artificial influences on trends over time, systematically assess how scholars use the data, and offer ideas for best practices for analysis.

The artifactual structure of lobbying disclosure data

Lobbying disclosures contain a great deal of information, but the report format makes deciphering the data confusing to the uninitiated. This confusion is an artifact of the original paper LD-1 forms that organizations filed to register themselves in the system and the semiannual (and later quarterly) LD-2 forms that registrants use to report lobbying activities.[2] The US House and Senate maintain separate depositories of both LD-1 registrations and LD-2 reports, though most registrants now file identical reports to both chambers through a single electronic submission.

Nested lobbying disclosure format

The original paper LD-2 reports had a top cover sheet that identified the registrant and client, listed contact information, and reported the aggregate amount of money spent or income earned, as well as the accounting method used to calculate that amount (Clemens 2014). Following the cover sheet, registrants then add one or more sheets for every pre-defined LDA Issue Area in which they were active.[3] The

[2] HLOGA added a new campaign donation reporting requirement for registered lobbyists. These reports do not disclose actual lobbying behavior, but rather donation behavior among a small percentage of lobbyists (Koger and Victor 2009; Victor and Koger 2016; McKay 2018).

[3] The original list included 72 issue areas; eight issue areas were added over time. See LaPira et al. (2014) for an extended discussion of how many issue areas overlap substantively and how the list is inexhaustive.

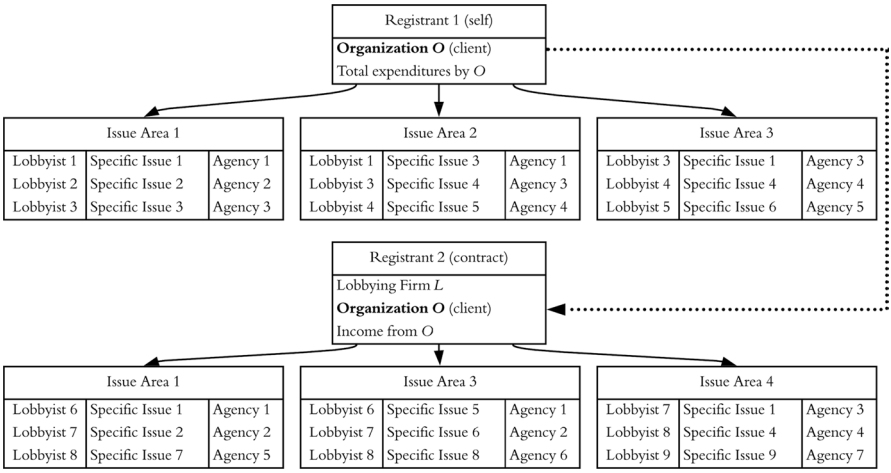

Fig. 1 Hypothetical Lobbying disclosures by an organization

issue area pages include names and "covered official" status[4] on lobbyists assigned to the Issue Area, an open-ended field briefly describing specific issues they advocated, and federal "agencies" contacted.[5] Since the number of issue areas varies for each interest group in each reporting period, it made sense for filers to simply add as many paper pages for as many issue areas they needed. When SOPR changed from paper to a digital system, they kept this page format as is even though doing so is inefficient from a database architecture perspective. The path-dependent consequence of this seemingly arbitrary decision is that reports continue to contain information in nested format. As we diagram in Fig. 1, information about lobbyists' identities and congressional bills the interest groups advocate, if mentioned at all,[6] are nested within the superordinate information on how much money is spent.

At first glance, lobbying disclosures appear to permit researchers to answer straightforward questions like "how much a lobbyist earned from tobacco interests?" or "do proponents or opponents spend more money on gun control?" Yet, this nested

[4] See "Lobbying Disclosure Act Guidance" for definitions. If a given lobbyist's covered official status was reported either in the original LD-1 or any previous LD-2 report for a given client, the status does not need to be reported again. This peculiar practice significantly undercounts revolving door lobbyists (LaPira and Thomas 2014, 2017).

[5] The US House and the US Senate are considered "agencies" under the LDA, which hides the actual members, committees, or party leaders and staff who are actually "contacted" by lobbyists.

[6] Prior to 2005, it was very rare for registrants to mention specific bills by standard House and Senate bill numbers. For the most part, bill numbers were used to identify legislation only when they were used colloquially by lawmakers, staff, lobbyists, and journalists. Otherwise, common titles or acronyms were used more frequently, or registrants simply described bills as "transportation funding" or "DoD reauthorization." The House Clerk began strongly urging filers to use bill numbers over time. No formal change to official guidance was made, but we do observe a dramatic rise in the number of bill number mentions beginning in 2006, before HLOGA passed.

format makes it impossible to answer these questions since expenditures are not itemized by lobbyist or issue.

Therefore, information nested within each report may be redundant. That is, a single quarterly report may indicate that the same lobbyist lobbied on the same bill in two different issue areas. This is, of course, artificial. When an actual lobbyist asks an actual Senator to vote for or against an actual bill, they do not ask twice in the same breath.

Complex lobbying disclosure reporting practices

To complicate matters further, the LDA requires entities to register and file under three separate conditions. First, if a registrant *spends* more than $5000 per quarter on lobbying, they report their aggregate expenditures (including money spent on lobbying firm consulting contracts) and all of their internal activities as if they are their own client. Typically, the organization filing under this condition has an office and full-time employees physically located in the Washington, DC metropolitan area. Second, if a consulting firm earns more than $5000 per quarter from a single client, they must report the activities on behalf of the client. The most common arrangement is a Washington lobbying or law firm that has multiple clients; they file separate LD-2 reports for each client in each reporting period. The third and least common practice is when a consulting firm exclusively engages in lobbying activity on behalf of a client, and the client organization does not engage in any in-house "lobbying activities" other than entering into the agreement with the firm. Since entering into a contract is itself a *de minimis* "lobbying activity," the organization does not file its own LD-2. So, for organizations filing exclusively under the third condition, their contract firms' lobbying income equals their total expenditures, which are not actually ever reported.

The original LD-2 forms, however, do not indicate under which conditions it is being filed. Fortunately, *OpenSecrets.org* deduces it when it cleans and codes the raw data, though users need to know how to query the bulk data to not (a) double-count a firm's income from a client as an expenditure or (b) genuinely count a firm's income as the client's only expenditures, as in the third condition.

The most common practice is for a single client organization and one or more of its consulting lobbying firms to file under the first and second conditions. While this method makes it easier to calculate overall expenditures, doing so necessarily reports redundant lobbying activity information. In other words, when information from a single client's many LD-2 s is aggregated to the organization level, it may actually make perfect sense to double-count lobbying *activities* even after taking careful steps to avoid double-counting lobbying *expenditures*. In Fig. 1, we show how subordinate data has a one-to-many relationship with a client across multiple reports filed on its behalf.

This diagram shows the nested and duplicative structure of information reported by hypothetical Organization *O* both as a self-filing registrant (top) and as a client of a second registrant (bottom), Lobbying Firm *L*. The organization reports its total expenditures on both its in-house expenses (e.g., overhead, compensation for

government affairs employees, travel and accommodations for visits to the Capitol, etc.) and its payments to consultants, in this case only one firm *L*. Organization *O* reports lobbying in three issue areas, employing five in-house lobbyists, advocating six specific issues, and contacting five agencies.

Lobbying firm *L* deployed four of its employees as additional contract lobbyists on Organization *O*'s behalf. Three of these lobbyists were also active in two issue areas that *O*'s in-house lobbyists were (Issue Areas 1 and 3), and three lobbyists were active on a completely new one, Issue Area 4. These contract lobbyists also lobbied on three specific issues not advocated by any of *O*'s in-house lobbyists. Not only is this hypothetical case complex, it also reveals redundant information, such as the fact that Specific Issue 4 was reported under two issue areas by in-house lobbyists and a third issue area by contract lobbyists. Redundant information like this can be used to understand the relative salience of particular specific issues. Specific Issue 4 is probably relatively more important to *O* than Specific Issue 9, which is mentioned only once across all six issue area reports.

These kinds of arrangements are not at all unusual, since some lobbyists specialize in substantive policy or in the relationships they have with particular lawmakers or agencies. At the same time, this particular hypothetical is much simpler than actual reports where a client uses dozens of lobbyists to engage in multiple activities. Moreover, multiply these complexities by the fact that all of these reports are filed semiannually (between 1996 and 2007) or quarterly (2008 to present) by tens of thousands of organizations lobbying at any given time.

With this complexity come great opportunities for creative measurement and identification strategies, since collectively the data allow for the construction of pooled time series, unbalanced panels, or relational networks now covering more than two decades of lobbying. Yet, the data themselves are characterized by several artificial characteristics, some of which can be traced to the 2007 HLOGA and other unclear or undocumented causes. In other words, there are disturbances in trends over time that have nothing to do with any theory about interest group politics, but need to be considered nonetheless.

Annualized trends in LDA reporting, 1998–2018

The CRP lobbying database contains 1000,336 verified LD-2 lobbying disclosure reports filed between 1998 and 2018.[7] Within these reports, a total of 47,555 unique organizations report lobbying activity at least once during the 21-year observation window.[8] Table 1 reports the ten most frequently occurring clients over time,

[7] 2018 was the most recent full reporting year at the time of writing. Note that CRP identifies the most recent report where amendments replace previously filed reports. We exclude reports replaced by amended filings. .

[8] CRP resolves organizational acquisitions, mergers, and rebrandings at the two-year federal election cycle level. That is, if Corporation *X* and Corporation *Y* merge to become Organization *Z* in 2016, lobbying and campaign finance reports for both *X* and *Y* are artificially attributed *Z* in 2015, even though it may not have existed.

Table 1 Client Reports, 1998–2018

Client	Reports
Pharmaceutical Research & Manufacturers of America	2189
Verizon Communications	1869
Lockheed Martin	1852
Comcast Corp	1726
Amgen Inc	1531
AT&T Inc	1484
General Dynamics	1413
Microsoft Corp	1324
Edison Electric Institute	1230
Boeing Co	1143
Overall N	47,555
Overall Mean (s.d.)/Median	21.0 (49.82)/7.0

LDA reports includes both semiannual and quarterly reports since 1998

aggregating both semiannual and quarterly reports since 1998. PhRMA (2189), Verizon (1869), and Lockheed Martin (1,1852) report most often, suggesting they retained a large number of consulting firms, hired a large number of lobbyists, and advocated in many issue areas. These data are highly skewed (Drutman et al. 2019), with a mean of 21.03 and median of 7 reports per client. If a client were to employ one in-house lobbyist or only one contract firm for 64 possible reporting periods (10 semiannual periods and 44 quarterly periods), they would occur 64 times at this level of analysis. In the case of PhRMA, the organization's 2189 reports include 64 periods of in-house lobbying, 64 periods of reports under contract with the major law firm Covington & Burling, and 2056 reports filed by 156 additional firms.

To address the nested and complex structure of the reports, data are typically reshaped at different levels of analysis by joining select data files available separately from the *OpenSecrets.org*.[9] Figure 2 plots the number of reports filed over time as well as the number of unique clients, total spending, and multi-client contract lobbying firms that register as agents of their clients. Since forms were filed semiannually prior to 2007 and quarterly from 2008 onward, we aggregate all reports within each year. Doing so is also consistent with common practice in lobbying consulting contracts, which are typically written to match the calendar year time frame and rarely, if ever, span across Congresses.

In the first year of the series, 19,882 lobbying disclosure reports were filed. This number increased steadily to a peak of 38,739 in 2007, and nearly doubled in 2008.

[9] CRP bulk lobbying files are divided into various "tables," each with their own data dictionary. Our analyses join these tables by the relevant unit of analysis. Typically, the table "lob_lobbying" is the primary object that contains one row per lobbying report. It can be joined to other tables to construct a dataset containing nested information such as issue areas ("lob_issue"), lobbyists ("lob_lobbyist"), and agencies ("lob_agency") as needed.

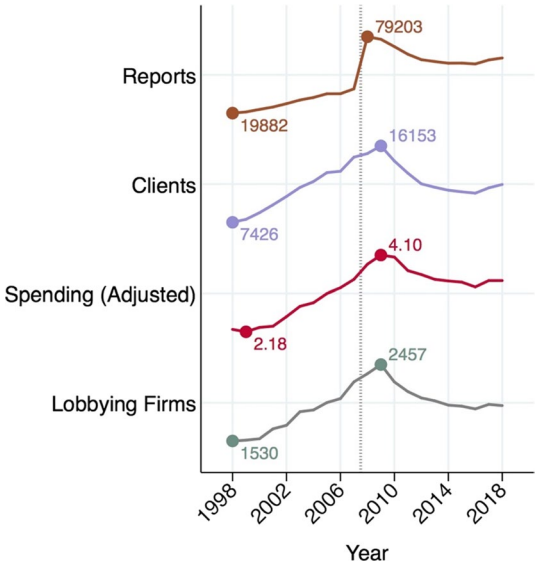

Fig. 2 Trends in LDA Reports, 1998–2008. Note: Counts are of non-missing observations for the 1998–2018 time span. Data are aggregated at the yearly level across semiannual (≤2007) and quarterly (>2007) LDA reports. Clients and firms are calculated as the unique number of each occurring in a given year. Spending is USD adjusted to 2018 dollars for inflation using the Consumer Price Index (CPI-U-RS, US Census), in billions across reports filed in a given year

While it appears that lobbying grew dramatically from 2007 to 2008, this shift is a consequence of increased disclosure frequency, not increased lobbying activity. Since the high of 79,203 in 2008, the number of reports filed decrease rapidly, ranging between 62,000 and 58,000 from 2012-2018. These declines are attributed to more conservative LDA compliance stemming from increased HLOGA reporting requirements and increased journalistic and academic scrutiny of the reports (LaPira 2016; Thomas and LaPira 2017; LaPira and Thomas 2017).

The other measures shown in Fig. 2—clients, firms, and total spending (adjusted for inflation)—are independent of increases in the volume due to quarterly filings. Each series exhibits a similar trend: steady increases between 1998 and 2009, a high in 2009, and a rapid decline through 2012. They then hold flat from 2012 to 2017 with slight increases in 2018.

Aggregating all reports to the client level yields the mean number of lobbyists, agencies mentioned, issue areas, and specific issues listed on lobbying disclosure forms over time.[10] Given the LD-2 report-level structure of the lobbying data, there are two methods by which researchers can count nested information. The first is to aggregate across reports filed each year, which produces counts that measure repeated and non-exclusive occurrences of lobbyists, agencies, and other measures. The second method is to tabulate unique occurrences of each value across clients.

This latter approach compensates for measurement error caused by the sharp increase in reporting frequency between 2007 and 2008, but does ignore potentially useful information on the volume of event counts of issue mentions, lobbyist mentions, and so on. For example, if a client's reports in a given year

[10] We do this by separately joining the "lob_lobbying" table to each table with subordinate information (e.g., "lob_lobbyist") and tabulating aggregated counts and means by client across each year.

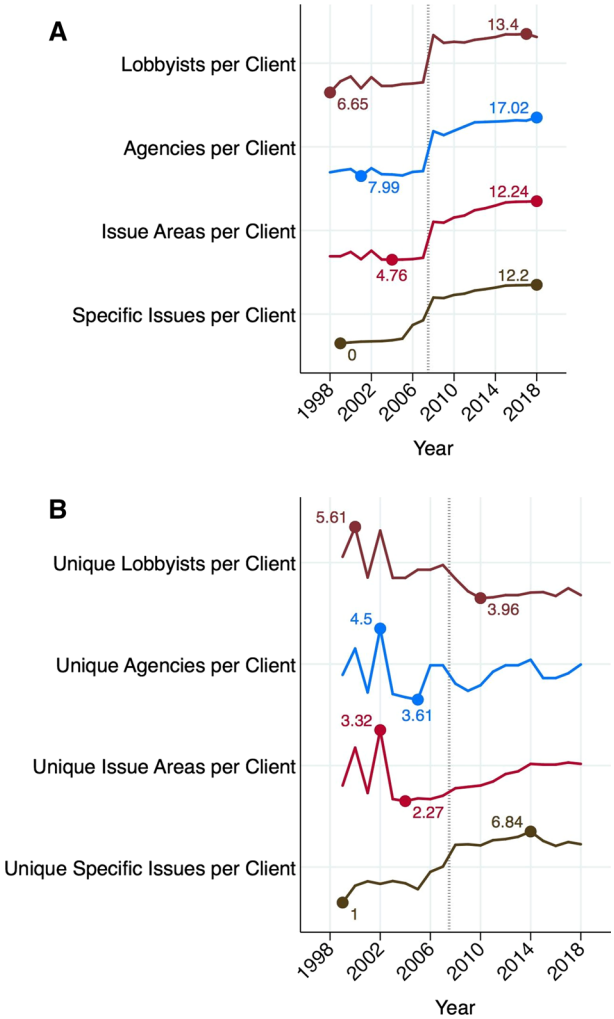

Fig. 3 **a** Trends in Nested Information by Clients (Means of Aggregated Observations), 1998–2008. Note: Measures are calculated as aggregated means across clients active in a given year. Lobbyists, agencies, issues, and specific issues may be repeated across reports for a given client. Agencies includes mentions of either chamber of the US Congress. **b** Trends in Nested Information by Clients (Means of Unique Observations), 1998–2008. Note: Measures are calculated as means across clients active in a given year. Measures are tabulated as unique mentions of individual lobbyists, agencies, issues, and specific issues. Agencies includes mentions of either chamber of the US Congress

mentioned the Federal Aviation Administration as a targeted "agency" on eight quarterly forms in 2018 (four filed by the client disclosing in-house lobbying and four filed by one contract lobbying firm), the first method would result in a count of eight FAA agency-mentioned events and the second would result in one count of the organization contacting the FAA.

We plot the mean number of lobbyists, agencies, issue areas, and specific issues using both methods in Fig. 3a, b. Figure 3 shows that each series exhibits a similar pattern: largely flat counts prior to HLOGA, a sharp increase in 2008 followed by an immediate decline, and a slowly increasing trend through 2018. This artificial increase illustrates how HLOGA presents significant issues for time series and panel analyses, despite the growing number of years of available data. If the research question demands aggregating data at the report level, then we recommend separating pre- and post-HLOGA reports into two separate, non-parallel series.

In Fig. 3b, the count of lobbyist, agency, issue area, and specific issue events present undisrupted trends. The first three series are subject to fluctuation between 1998 and 2003 unrelated to changes to the law several years after. The trends stabilize and the series diverge from 2003 to 2018. The number of lobbyists per client increases slightly through 2007, declines until 2010 and holds at roundly 4 through 2018. The number of agencies oscillates between 3.6 in 2005 to 2004 through the end of the series.

Aggregated across all reports, the US House and Senate account for 29.6% and 29.4% of the nearly 3 million mentions of agencies, with 250 other government offices and departments listed. When tabulated as unique occurrences across clients, the US House and Senate each account for an identical 18.37% of 230,300 observations, suggesting a distinction without a difference.

The number of issue areas shown in Fig. 3b increases steadily from a low of 2.3 in 2004 up to 3 in 2018. This increase may in part be the result of additional issue areas available to filers to categorize their activity—up from 72 to 80 since 1998. Across the full time period, the issues of Federal Budget & Appropriations, Taxes, Health Issues, Defense, and Transportation account for roughly a third of all issue area activity.

Unique mentions of specific issues increase steadily over time. Over 800,000 specific issues are listed in LDA reports since 1998. They range from general descriptions of activity such as "monitor trade legislation," "copyright reform," or "appropriations" to highly specific and enumerated lists of bill names and numbers, and even specific perfecting amendments or subordinate policies contained in proposed legislation. Though CRP does parse bill numbers and names from these specific issue fields beginning in 2005, researchers should be extremely cautious in their use. Given the lack of unique bill id numbers that span Congressional sessions, and varied detail in disclosure in the specific issue field, the potential for measurement error in terms of matching listed bills to bill content is high.[11]

[11] Dusso (2010), Grossmann and Pyle (2013), LaPira and Thomas (2017), Brasher and Britt (2014), and You (2017) analyze bill mentions from CRP data, but indicate significant manual data cleaning effort to minimize obvious sources of measurement error.

The scholarly use of LDA data

Interest group scholars have made good use of LDA data in the quarter century they have been available. Yet we argue that the interest group scholarly community has, ironically, suffered from a collective action failure to exploit the data to its full potential. Here, we identify 85 peer-reviewed articles, chapters, and books that rely on LDA data in a variety of ways to uncover two general patterns.

First, research that uses only LDA data tends to focus on selected cases—typically major corporations in the *Fortune* 500 or 1000, specific organizations like the National Rifle Association, or particular policy issues such as taxes or the environment. This work offers insight into interest group behavior, but inherent biases from selection on cases or observables make these insights difficult to generalize across the full population. Second, research designed to be more generalizable tends to join LDA data with information from other large data sources. This kind of work lends itself to generalizability, but scholars tend to create single-use connections to secondary data sources to test specific hypotheses. These synthetic "LDA + other" datasets are useful in isolation and for replication, but there is little effort to pool them together as a common good resource. That is, despite the enormously rich LDA data, scholars continue to reinvent the wheel to put the data to good—albeit one-time use.

Analysis of LDA research

We conducted a comprehensive search to identify research using LDA data. We initially search library catalogs and public sources like Google Scholar. We then add sources based on our familiarity with the literature and extensive literature reviews (de Figueiredo and Brian 2014; Hojnacki et al. 2012), searching major academic presses and journals, and browsing every issue of *Interest Groups & Advocacy* journal.[12]

We report the coverage (in years) of lobbying data used, the unit(s) of analysis, the authors' sampling process, and the scope of data use, and any datasets that are merged with LDA data in Table 2. In terms of coverage, 17 studies (20%) leverage only one year of data and the overwhelming majority use multiple years. For those spanning multiple years, it is common for researchers to match coverage to selection needs (e.g., the years in which a linked dataset is observed). On average, the studies in our analysis span 6.7 years of LDA data (s.d. = 5.4). The median start year is 1999 and the median end year is 2008.

Units of analyses vary quite a bit. We find that 51 (60.0%) of studies use the client level of analysis. Another 16 (18.8%) use the lobbyist, focusing on revenue, partisanship, or revolving door status. Three studies examine lobbying activity by clients aggregated to the level the economic industry.[13] Another four studies (4.7%) analyze

[12] See the Online appendix for more detail on the search process, coding scheme, and a comprehensive list of research works.

[13] Researchers studying industry-level activity typically focus on business clients and used linked datasets to aggregate to NAICS industries. CRP also adds a hierarchical industry classification to client-level

Table 2 Analysis of research using LDA data

Coverage	
Mean Years (s.d.)	6.7 (5.4)
Unit of Analysis	
Client	51 (60.0%)
Lobbyist	16 (18.8%)
Industry	3 (3.5%)
Issue Area	4 (4.7%)
Bill	3 (3.5%)
Lobbying Firm	1 (1.2%)
Multiple	4 (4.7%)
Other	3 (3.5%)
Links to datasets	
Compustat or Other Financial	13 (15.3%)
CRP Campaign Finance/PAC Contributions	5 (5.9%)
Policy Agendas Project	4 (4.7%)
Lobbyists.info/Washington Reps.	3 (3.5%)
Other	34 (40.0%)
Scope of Inquiry	
Full population	17 (20.0%)
Random sample	12 (14.1%)
Selection	42 (49.4%)
Single case	6 (7.1%)
N	85

the data at the issue area level, three at the bill level (3.5%), and one (1.2%) at the level of lobbying firms. Others simultaneously draw on multiple levels (e.g., lobbyists and lobbying firms), or link lobbying activity to a study of another unit of analysis (e.g., policy domains or agency rules).

Scholars merge a wide range of outside data sources with lobbying disclosure data. The most common was the corporate financial information (e.g., revenue) drawn from *Compustat* or *S&P Corporate IQ* with 13 (15.3%). Nearly all studies of businesses as lobbying clients draw on these data sources for important covariates. Five studies (5.9%) compared lobbying activity to campaign finance activity or linked individual lobbyists with specific donations, commonly drawn from CRP either at the organization or lobbyist level of analysis.

The various datasets published by the Policy Agendas Project (Baumgartner, Jones, and Wilkerson 2002; Jones and Baumgartner 2005) are linked to LDA data in four instances (4.7%).[14] These include information on *Gallup's* Most Important

Footnote 13 (continued)
data, though this schema is not easily converted into the popular NAICS categories (see https://www.opensecrets.org/federal-lobbying/industries?cycle=a).

[14] See https://www.comparativeagendas.net/us.

Problem dataset or measures of government attention in the form of Congressional hearings. Three studies (3.5%) use information from *Washington Representatives*. An additional 34 (40%) studies merge LDA data to other datasets that range from interest group scorecards of Members of Congress, *Maplight* bill position information, characteristics of US cities, and US state-level lobbying activity.

The scope of analyses is also highly varied. Twelve studies (14.1%) draw random samples of clients, lobbyists, or issue areas, while 17 (20.0%) analyze the full population of clients, lobbyists, or issues in a given time frame. Nearly half of the 85 studies (42, 49.4%) limit their scope to a selection driven by the research question. Selections span clients meeting certain criteria—such as companies listed in the *S&P* 500, midsize American cities, certain institutions of higher education, or electric utilities. The remaining 6 (7.1%) focus on single cases such as the lobbying of one client or lobbying surrounding high-profile legislation like Dodd-Frank or the Affordable Care Act.

Selecting cases may be advantageous for given questions or may be necessary for feasibility, though generally scholars fail to recognize their selection biases. More importantly, creating datasets on selections like these generates a tragedy of the commons. For example, there exists no common, easily accessible, and comprehensive crosswalk between the LDA and *Compustat* even though 15% of the studies merge these data sources.[15] Similarly, any new study of lobbying clients must generate supplemental information (e.g., group type) independently, despite these measures being routinely collected among scholars of organized interests.

The future of LDA data

The longitudinal and cross-sectional depth of the LDA data is underused. This may be due to the complex, nested structure of the data and the high start-up costs involved in merging lobbying information with other sources of data. To aid in this effort, we make available a number of tables drawn the CRP lobbying database.[16] Most importantly, we generate up-to-date datasets at a variety of levels of analysis that aggregate the nested information across the entire CRP observation period (1998-2018), and plan to do so for each subsequent quarterly reporting period.[17] The population of clients, lobbying firms, lobbyists, and issue areas presents new opportunities for generalizable findings about lobbying behavior beyond purposive or convenience samples.

Our analysis of the 85 articles and books that leverage LDA data also suggests the need for a new "data commons" to support the study of organized interests. We

[15] Note that the LobbyView project makes great strides in this direction (see https://www.lobbyview.org/#!/), though the data are available only in small quantities through a programming interface.

[16] Available at author website. The replication data would be in addition to CRP data, recompiled and reshaped as we do in this article.

[17] We also make available for replication the code and longitudinal dataset compiled to produce Figs. 2 and 3 above at author website.

suggest that significant analytical progress could be made through centralizing client-level information in a single location, organized with identifiers for organizations, lobbying firms, lobbyists, and bills per Congress. The lobbying data commons would allow users merge relevant datasets to meet their needs, thus avoiding the need to duplicate efforts with each new study.

As the LDA turns 25, we look forward to reflecting on the database at 50. Ultimately, we aspire to go beyond relying simply on a lobbying disclosure data commons, but rather a universal sampling frame of all organizations engaged in American politics. We acknowledge that the lobbying disclosure database itself is a narrow selection of groups that engage in some form of advocacy at the US national level only. Most organizations that seek to influence legislation, regulations, or legal interpretations probably do not report those activities if they do not trigger LDA's narrow reporting requirements (Thomas and LaPira 2017), yet researchers have compiled extensive information on organizations active in other venues (e.g., submission of amicus curiae briefs, Box-Steffensmeier and Christenson 2014). Achieving this goal would be an enormous challenge, but would create new and exciting opportunities for the scholarly community to learn more about how well-organized interests influence who gets what, when, and how.

Compliance with ethical standards

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

Baumgartner, Frank R., Bryan D. Jones, and John D. Wilkerson. 2002. Studying Policy Dynamics. In *Policy Dynamics*, edited by Frank R Baumgartner and Bryan D Jones, First. University of Chicago Press.

Baumgartner, Frank R., and Beth L. Leech. 1998. *Basic Interests: The Importance of Groups in Politics and in Political Science*. Princeton: Princeton University Press.

Baumgartner, Frank R., and Beth L. Leech. 1999. Studying Interest Groups Using Lobbying Disclosure Reports. *Vox Pop* 17 (3): 1–3.

Baumgartner, Frank R., and Beth L. Leech. 2001. Issue Niches and Policy Bandwagons: Patterns of Interest Group Involvement in National Politics. *Journal of Politics* 63: 1191–1213.

Box-Steffensmeier, Janet M., and Dino P. Christenson. 2014. The Evolution and Formation of Amicus Curiae Networks. *Social Networks* 36: 82–96.

Brasher, Holly, and Jason Britt. 2014. Understanding the Influence of Lobbying in the U.S. Congress: Preferences, Networks, Money, and Bills. In *New Directions in Interest Group Politics*, edited by Matt Grossmann, 185–203. New York: Routledge.

Clemens, Austin C. 2014. All Politics Is Local, but Lobbying Is Federal and Local: The Validity of LDA Data. *Business and Politics* 16 (2): 267–289.

Drutman, Lee, Matt Grossmann, and Timothy LaPira. 2019. The Interest Group Top Tier. In *Can America Govern Itself?* ed. Frances E. Lee, and Nolan McCarty, 45–68. New York: Cambridge University Press.

Dusso, Aaron. 2010. Legislation, Political Context, and Interest Group Behavior. *Political Research Quarterly* 63: 55–67.

de Figueiredo, John M., and K.Richter. Brian. 2014. Advancing the Empirical Research on Lobbying. *Annual Review of Political Science* 17: 163–185.

Gray, Virginia, and David Lowery. 1996. *The Population Ecology of Interest Representation*. Ann Arbor: University of Michigan Press.

Grossmann, Matt, and Kurt Pyle. 2013. Lobbying and Congressional Bill Advancement. *Interest Groups & Advocacy* 2 (1): 91–111.

Heinz, John P., Edward O. Laumann, Robert L. Nelson, and Robert H. Salisbury. 1993. *The Hollow Core: Private Interests in National Policymaking*. Cambridge: Harvard University Press.

Hojnacki, Marie, David C. Kimball, Frank R. Baumgartner, Jeffrey M. Berry, and Beth L. Leech. 2012. Studying Organizational Advocacy and Influence: Reexamining Interest Group Research. *Annual Review of Political Science* 15: 379–399.

Jones, Bryan D., and Frank R. Baumgartner. 2005. *The Politics of Attention: How Government Prioritizes Problems*. Chicago: University of Chicago Press.

Koger, Gregory, and Jennifer Nicoll Victor. 2009. Polarized Agents: Campaign Contributions by Lobbyists. *PS: Political Science and Politics* 42 (3): 485–88.

LaPira, Timothy M. 2016. Lobbying in the Shadows: How Private Interests Hide From Public Scrutiny and Why That Matters. In *Interest Group Politics*, edited by Allan J. Cigler and Burdett A. Loomis, 9th ed., 224–48. Washington: CQ Press.

LaPira, Timothy M., and Herschel F. Thomas. 2014. The Revolving Door and Interest Representation. *Interest Groups and Advocacy* 3 (1): 1–26.

LaPira, Timothy M., and Herschel F. Thomas. 2017. *Revolving Door Lobbying: Public Service, Private Influence, and the Unequal Representation of Interests*. Lawrence: University Press of Kansas.

LaPira, Timothy M., Herschel F. Thomas, and Frank R. Baumgartner. 2014. The Two Worlds of Lobbying: Washington Lobbyists in the Core and on the Periphery. *Interest Groups and Advocacy* 3: 219–224.

McKay, Amy. 2018. What Do Campaign Contributions Buy? Lobbyists' Strategic Giving. *Interest Groups and Advocacy* 7 (1): 1–18.

Schlozman, Kay L. 1984. What Accent the Heavenly Chorus? Political Equality and the American Pressure System. *The Journal of Politics* 46: 1006–1032.

Schlozman, Kay Lehman, and John T. Tierney. 1983. More of the Same: Washington Pressure Group Activity in a Decade of Change. *Journal of Politics* 45: 351–377.

Schlozman, Kay Lehman, and John T. Tierney. 1986. *Organized Interests and American Democracy*. New York: Harper and Row.

Thomas, Herschel F., and Timothy M. LaPira. 2017. How Many Lobbyists Are in Washington?: Shadow Lobbying and the Gray Market for Policy Advocacy. *Interest Groups and Advocacy* 6 (3): 199–214.

Victor, Jennifer Nicoll, and Gregory Koger. 2016. Financing Friends: How Lobbyists Create a Web of Relationships among Members of Congress. *Interest Groups and Advocacy* 5 (3): 224–262.

Walker, Jack L. 1983. The Origins and Maintenance of Interest Groups in America. *American Political Science Review* 77: 390–406.

You, Hye Young. 2017. Ex Post Lobbying. *The Journal of Politics* 79 (4): 1162–1176.

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